Data Science

CSE558

Recap

- Universal Hashing

$$\mathbf{Pr}_{h \in \mathcal{H}}\left(h(x) == h(y)\right) = \frac{1}{m}$$

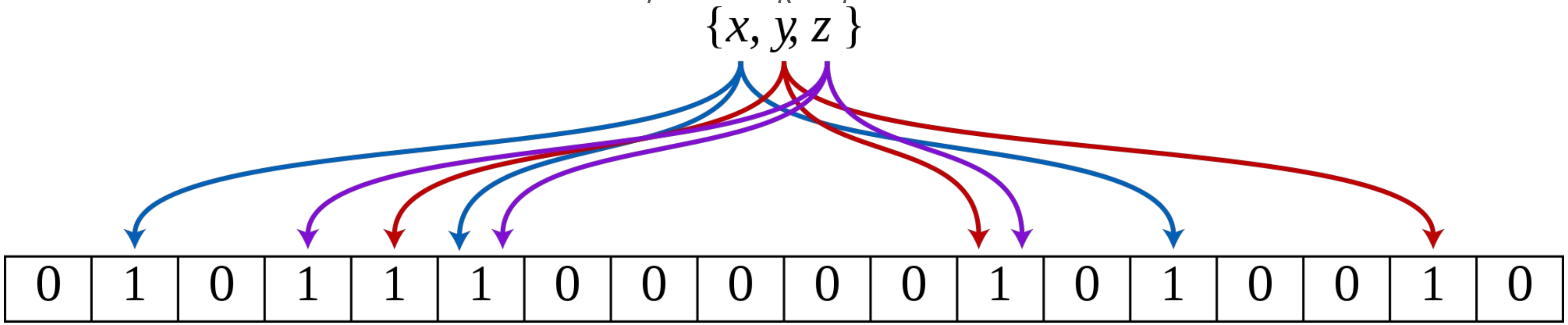
$$\mathcal{H} = \{h_a(x) = (ax \mod p) \mod m, a \in [1, \ldots, p-1]$$

- Bloom Filter

Bloom Filter (Bloom, 1970)

Bit array of size m

k, independent hash functions $h_1, \ldots, h_k$. h_i: $|U| \to [m]$



Initialize: B[i] = 0 for every $1 \leq i \leq m$

Insert(B,x): B[$h_j(x)$] = 1, for every $1 \leq j \leq k$

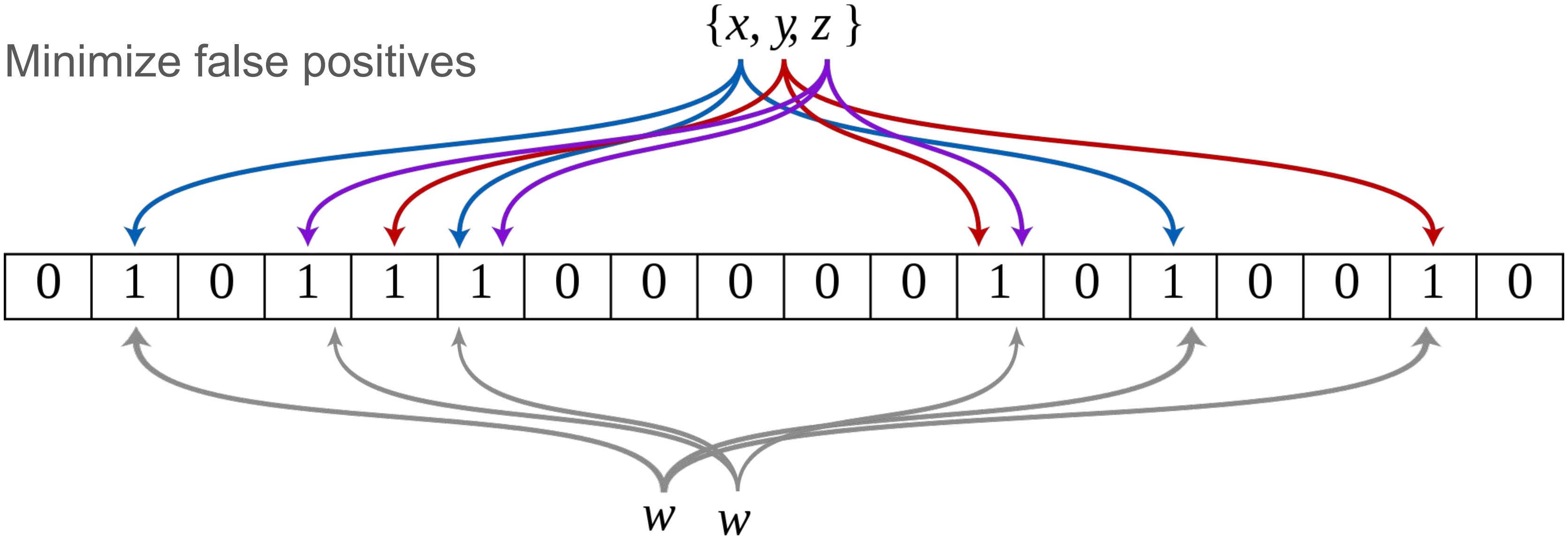
Query(B, w): If B[$h_1(w)$] $\wedge$ B[$h_3(w)$] $\wedge \ldots \wedge$ B[$h_k(w)$] = 1, then return true else false.

Bloom Filter (Bloom, 1970)

If w is present then it will return true.

When w not present then there might be false positive.

Minimize false positives



Applications

Cache Protocol (Akamai): Avoids caching one-hit-wonders. 75% of web pages are only requested once, so caching them only on the second request significantly improves performance.

Safe search: Stores bloom filter of malicious url (or i.p.)

Validate weak password: Stores weak or compromised passwords in the bloom filter.

Overview

- Distinct Elements

Streaming Data

Consider massive data arriving in streaming fashion.

- Internet traffic: 0.5B tweets, 10B searches etc
- Large datasets: Imagenet with 14M images.
- Sensor data: 15T per night from Telescope, Seismometer arrays for earthquake prediction, Traffic sensors.

Goal is

- Analyze & learn from the data.
- Use randomized compression (Bloom Filters)

Distinct Elements

Let $X_1, X_2, \ldots, X_n$ be a stream of elements. The task is to estimate the number of distinct elements.

<u>Applications:</u>
Distinct add visitors, Distinct number of searches, Distinct users

<u>Algo:</u>
Let h: U → [L] // a random hash function
z = 0
For i = 1, …, n
    z = max(z,zeros(h(X_i))
Return $2^{z+½}$

Distinct Elements

R, B, R, R, G, B, R, R, B, G, Y, B

B: 0110101

R: 1111010

G: 1000100

Y: 1011010

Distinct Elements (Analysis)

For any $r \in [L]$, $j \in U$, $R_{r,j}$: indicator of $zeros(h(X_j)) > r$

Let $Y_r = \Sigma_j R_{r,j}$ and z^* be the returned z value.

Properties: $z^* \geq r$, iff $Y_r > 0$ and $z^* < r$. iff $Y_r = 0$

Compute $E[Y_r]$ and $Var(Y_r)$

Compute $Pr(Y_r > 0) = Pr(Y_r \geq 1)$

Compute $Pr(Y_r = 0) \leq Pr(|Y_r - E[Y_r]| \geq E[Y_r])$

Distinct Elements (Analysis)

Returned estimate: $n^* = 2^{z^*+½}$

Upper bound:
Let a be the smallest integer such that, $2^{a+½} \geq 4n$.

$\Pr(n^* \geq 4n) = \Pr(z^* \geq a) = \Pr(Y_a > 0)$

Lower bound:
Let b be the largest integer such that, $2^{b+½} \leq n/4$.

$\Pr(n^* \leq n/4) = \Pr(z^* \leq b) = \Pr(Y_{b+1} = 0)$

Median Trick: Boost success probability with the median.

Distinct Elements (Analysis)

Space Complexity:

$L = c \cdot \log(n)$ (why this is enough?)

$z = O(\log(\log(n)))$

Distinct Elements

Let X_1, X_2, …, X_n be a stream of elements. The task is to estimate the number of distinct elements.

<u>Algo2:</u>

Let h: U → [0,1] // a random real valued hash function

s = 1

For i = 1, …, n

    s = min(s,h(X_i))

Return 1/s - 1

Distinct Elements

What is E[s].

For 2 distinct points?

For 3 distinct points?

For k distinct points?

E[s] = 1/(d+1). Is E[1/s - 1] = d?

Distinct Elements

Let d be the #unique elements.

Bound $\Pr(|s - E[s]| \geq \varepsilon \cdot E[s])$
How useful is this?

Algo3:
Let $h_1, \ldots, h_k : U \to [0,1]$ // a random real valued hash function
$s_1 = \ldots = s_k = 1$
For $i = 1, \ldots, n$
    For $j = 1, \ldots, k$
        $s_j = \min(s_j, h_j(X_i))$
$s = (s_1 + \ldots + s_k)/k$
Return $1/s - 1$

Distinct Elements

Compute, k for $\Pr(|s - E[s]| \geq \varepsilon \cdot E[s]) \leq \delta$.

Medians of Means:

Let $k = \log(1/\delta)/\varepsilon^2$

1. Maintain a $1/\varepsilon^2$ groups each of size $\log(1/\delta)$.
2. Compute the mean of the n* in each groups
3. Compute the median of the means.

Space: O(klog(n))

Update: O(k)

Application

Consider two separate streams s_1 and s_2 that uses the same hash functions.

Compute

- $s_1 \cup s_2$
- $s_1 \cap s_2$.